No.	Status	Date
LNDE-RPT-01-037-DS-6-RA-103	Issued	2001-03-02

UT Thickness Data Sheet - 6-RA-103

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the rich amine header 6-RA-103 located within the ATU-100. The inspection was conducted by Grace Okafor to support the 5-year Class 1 survey requirements.

The technician is responsible for the accurate acquisition of thickness data and the identification of any readings falling below the minimum allowable thickness. All activities were performed in accordance with the site requirements for the Amine Treating Unit.

Technical Basis

The measurements were performed under the requirements of QSCP-NDT-001 Rev 0. The ultrasonic testing methodology and equipment calibration were executed in accordance with ASTM E797.

The data collected for this line is integrated into the comprehensive findings detailed in LNDE-RPT-01-037.

Line Specifications

The subject line, 6-RA-103, is constructed from A106 Gr.B with a nominal thickness of 10.97 mm and an outer diameter of 168.3 mm. The design pressure is 862 kPa(g) and the design temperature is 82 ?C.

The minimum allowable thickness (t-min) for this asset is 4.57 mm. Any measured value below this threshold is subject to the site non-conformance reporting process.

Analysis and Evaluation

Corrosion rates are determined by comparing the current readings against historical data, utilizing the larger of the long-term rate or the short-term rate over a minimum 3-year interval.

Remaining life is calculated based on the difference between the current thickness and the 4.57 mm t-min divided by the calculated rate. The subsequent inspection interval is set at half the remaining life, not exceeding the Class 1 maximum of 5 years.

Thickness readings, measured 2001-03-02

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	1996-03-08	7.12	6.82	0.060	0.060	>50	2006-03-02	
6-SG-101-02	7.11	3.56	1996-03-08	7.05	6.71	0.068	0.068	46.2	2006-03-02	
6-SG-101-03	7.11	3.56	1996-03-08	7.08	6.74	0.068	0.068	46.6	2006-03-02	
6-SG-101-04	7.11	3.56	1996-03-08	7.09	6.73	0.072	0.072	43.9	2006-03-02	
6-SG-101-05	7.11	3.56	1996-03-08	7.14	6.68	0.092	0.092	33.8	2006-03-02	
6-SG-101-06	7.11	3.56	1996-03-08	7.07	6.87	0.040	0.040	>50	2006-03-02	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	1996-03-08	7.14	6.72	0.084	0.084	37.5	2006-03-02	
6-SG-102-02	7.11	3.56	1996-03-08	7.11	6.72	0.078	0.078	40.4	2006-03-02	
6-SG-102-03	7.11	3.56	1996-03-08	7.04	6.69	0.070	0.070	44.6	2006-03-02	
6-SG-102-04	7.11	3.56	1996-03-08	6.99	6.64	0.070	0.070	43.8	2006-03-02	
6-SG-102-05	7.11	3.56	1996-03-08	7.13	6.93	0.040	0.040	>50	2006-03-02	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	1996-03-08	8.52	8.39	0.026	0.026	>50	2011-03-02	
4-RA-101-02	8.56	3.81	1996-03-08	8.54	8.45	0.018	0.018	>50	2011-03-02	
4-RA-101-03	8.56	3.81	1996-03-08	8.61	8.42	0.038	0.038	>50	2011-03-02	
4-RA-101-04	8.56	3.81	1996-03-08	8.63	8.37	0.052	0.052	>50	2011-03-02	
4-RA-101-05	8.56	3.81	1996-03-08	8.60	8.42	0.036	0.036	>50	2011-03-02	
4-RA-101-07	8.56	3.81	1996-03-08	8.60	8.51	0.018	0.018	>50	2011-03-02	
4-RA-101-08	8.56	3.81	1996-03-08	8.68	8.30	0.076	0.076	>50	2011-03-02	
4-RA-101-09	8.56	3.81	1996-03-08	8.57	8.43	0.028	0.028	>50	2011-03-02	
4-RA-101-10	8.56	3.81	1996-03-08	8.67	8.48	0.038	0.038	>50	2011-03-02	
4-RA-101-11	8.56	3.81	1996-03-08	8.68	8.39	0.058	0.058	>50	2011-03-02	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	1996-03-08	8.54	8.48	0.012	0.012	>50	2011-03-02	
4-RA-102-02	8.56	3.81	1996-03-08	8.57	8.50	0.014	0.014	>50	2011-03-02	
4-RA-102-03	8.56	3.81	1996-03-08	8.56	8.37	0.038	0.038	>50	2011-03-02	
4-RA-102-04	8.56	3.81	1996-03-08	8.54	8.35	0.038	0.038	>50	2011-03-02	
4-RA-102-05	8.56	3.81	1996-03-08	8.55	8.36	0.038	0.038	>50	2011-03-02	
4-RA-102-06	8.56	3.81	1996-03-08	8.51	8.38	0.026	0.026	>50	2011-03-02	
4-RA-102-07	8.56	3.81	1996-03-08	8.58	8.49	0.018	0.018	>50	2011-03-02	
4-RA-102-08	8.56	3.81	1996-03-08	8.49	8.36	0.026	0.026	>50	2011-03-02	
4-RA-102-09	8.56	3.81	1996-03-08	8.55	8.35	0.040	0.040	>50	2011-03-02	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	1996-03-08	11.00	10.82	0.036	0.036	>50	2011-03-02	
6-RA-103-02	10.97	4.57	1996-03-08	11.09	10.69	0.080	0.080	>50	2011-03-02	
6-RA-103-03	10.97	4.57	1996-03-08	10.99	10.84	0.030	0.030	>50	2011-03-02	
6-RA-103-04	10.97	4.57	1996-03-08	10.98	10.59	0.078	0.078	>50	2011-03-02	
6-RA-103-05	10.97	4.57	1996-03-08	10.95	10.61	0.068	0.068	>50	2011-03-02	
6-RA-103-06	10.97	4.57	1996-03-08	11.01	10.86	0.030	0.030	>50	2011-03-02	
6-RA-103-07	10.97	4.57	1996-03-08	10.99	10.79	0.040	0.040	>50	2011-03-02	
6-RA-103-08	10.97	4.57	1996-03-08	10.97	10.83	0.028	0.028	>50	2011-03-02	
6-RA-103-09	10.97	4.57	1996-03-08	11.01	10.85	0.032	0.032	>50	2011-03-02	
6-RA-103-10	10.97	4.57	1996-03-08	11.02	10.78	0.048	0.048	>50	2011-03-02	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	1996-03-08	7.59	7.26	0.066	0.066	>50	2006-03-02	
3-RV-101-02	7.62	3.56	1996-03-08	7.57	7.31	0.052	0.052	>50	2006-03-02	
3-RV-101-03	7.62	3.56	1996-03-08	7.52	7.15	0.074	0.074	48.3	2006-03-02	
3-RV-101-04	7.62	3.56	1996-03-08	7.59	7.18	0.082	0.082	44.0	2006-03-02	
3-RV-101-05	7.62	3.56	1996-03-08	7.67	7.25	0.084	0.084	43.8	2006-03-02	
3-RV-101-06	7.62	3.56	1996-03-08	7.59	7.20	0.078	0.078	46.5	2006-03-02	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	1996-03-08	5.53	5.29	0.048	0.048	>50	2006-03-02	
2-RV-102-02	5.54	2.79	1996-03-08	5.59	5.12	0.094	0.094	24.7	2006-03-02	
2-RV-102-03	5.54	2.79	1996-03-08	5.48	5.24	0.048	0.048	>50	2006-03-02	
2-RV-102-04	5.54	2.79	1996-03-08	5.58	5.33	0.050	0.050	>50	2006-03-02	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	1996-03-08	8.51	8.40	0.022	0.022	>50	2011-03-02	
4-RA-104-02	8.56	3.81	1996-03-08	8.55	8.41	0.028	0.028	>50	2011-03-02	
4-RA-104-03	8.56	3.81	1996-03-08	8.54	8.46	0.016	0.016	>50	2011-03-02	
4-RA-104-04	8.56	3.81	1996-03-08	8.59	8.35	0.048	0.048	>50	2011-03-02	
4-RA-104-05	8.56	3.81	1996-03-08	8.60	8.29	0.062	0.062	>50	2011-03-02	
4-RA-104-06	8.56	3.81	1996-03-08	8.59	8.39	0.040	0.040	>50	2011-03-02	
4-RA-104-07	8.56	3.81	1996-03-08	8.59	8.29	0.060	0.060	>50	2011-03-02	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	2001-03-02	_____